



SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3
– Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q1 Student Management System [BL3 – Apply]

10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

CLASS Student

PRIVATE name : STRING

```
PRIVATE rollNo : INTEGER
```

PRIVATE marks : ARRAY OF FLOAT

PRIVATE average : FLOAT

PUBLIC METHOD acceptDetails()

OUTPUT "Enter Student Name : "

INPUT name

OUTPUT "Enter Roll Number : "

INPUT rollNo

10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
OUTPUT "Enter Roll Number : "  
INPUT rollNo  
  
OUTPUT "Enter number of subjects : "  
INPUT n  
  
DECLARE marks AS ARRAY OF SIZE n  
  
FOR i <- 0 TO n-1 DO  
    OUTPUT "Enter marks for subject ", i+1, ": "  
    INPUT marks[i]  
END FOR  
END METHOD  
  
PRIVATE METHOD calculateAverage( )  
    DECLARE sum <- 0  
  
    FOR EACH m IN marks DO  
        sum <- sum + m  
    END FOR  
  
    average <- sum / LENGTH(marks)
```

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Q10 ✓
10. Smart Home Device
Controller [BL4 – Analyze]
10 marks

```
        average <- sum / LENGTH(marks)
    END METHOD
```

```
    PUBLIC METHOD displayResult( )
        CALL calculateAverage( )
```

```
        OUTPUT "Student Name: ", name
        OUTPUT "Roll Number: ", rollNo
        OUTPUT "Average Marks: ", average
```

```
        IF average >= 50 THEN
            OUTPUT "Result: PASS"
        ELSE
            OUTPUT "Result: FAIL"
        END IF
    END METHOD
```

```
END CLASS
```

```
BEGIN
    DECLARE s AS Student
    CALL s.acceptDetails( )
    CALL s.displayResult( )
END
```